MIL-E-1/368E (See note 6)
28 November 1973
SUPERSEDING
MIL-E-1/368D
13 July 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 5865

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, bandpass, frequency range 5,395 to 5,905 MHz, peak power rating 300 kw

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb	Altitude
Unit:	kw	Vdc	feet
Maximum:	300	---	10,000
Minimum:	4	-700	---

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS:

Parameter:	tp1	tp2	prr	Ii	Incident power	σ'	F
Unit:	μs	μs	pps	μAdc	kw	---	MHz
Tolerance:	±0.15	±0.15	---	---	±10%	Max	---
Test condition 1:	1.0	0.50	1,000	100	4	1.1:1	F3
Test condition 2:	1.0	0.50	1,000	100	70	1.1:1	F3
Test condition 3:	1.0	---	1,000	100	4	1.1:1	F3
Test condition 4:	1.0	---	1,000	100	70	1.1:1	F3
Test condition 5:	1.0	---	1,000	100	300	1.1:1	F3

Frequency		
F	MHz	±
1	5,395	0.1%
2	5,450	0.1%
3	5,650	5 %
4	5,825	0.1%
5	5,905	0.1%

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4021	Degradation due to vibration	---	---		---	---	---	---
4452	Flat leakage power	---	1		pf	---	50	mw
4452	Spike leakage energy	---	1		Ws	---	0.25	erg
	Quality conformance inspection, part 1	1	---					
4473	Low-level VSWR	---	---	σ' = 1.05 max				
				F1	σ	---	1.9	---
				F2	σ	---	1.4	---
				F3 ±0.1%	σ	---	1.4	---
				F4	σ	---	1.4	---
				F5	σ	---	1.9	---
4416	Insertion loss	---	---	F = F3; Ii = 0	Li	---	0.8	dB
4421	Ignitor interaction	---	---	Ii = 100 μAdc	ΔLi	---	0.2	dB
4401	Ignitor ignition time	2	---	Ebb = -700 Vdc; R = 4.0 Meg ±1%	t	---	5	sec
4406	Ignitor voltage drop	---	---	Ebb = -700 Vdc; R = 4.0 Meg ±1%	Eid	200	400	Vdc
4452	Flat leakage power	---	2		pf	---	50	mw
4452	Spike leakage energy	---	2		Ws	---	0.25	erg
4494	Position of short	3	4		Distance	0.070	0.100	Inch
4488	Arc loss	---	3		Loss	---	0.80	dB
1027	Temperature cycling (nonoperating)	---	---		---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	5	---		---	---	---	---
4471	Recovery time	---	5		t	---	10.0	μs
	Quality conformance inspection, part 3							
---	Life test	4	5	Group C; Ebb = -700 Vdc; R = 4.0 Meg ±1%	t	500	---	hrs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end points:							
4471	Recovery time	---	5		t	---	20	μs
4452	Flat leakage power	---	2		pf	---	70	mw
4416	Insertion loss	---	---	F = F3; Ii = 0	Li	---	0.8	dB
4452	Spike leakage energy	---	2		Ws	---	0.3	erg
1027	Temperature cycling life test	---	---	Group C; 10 cycles	---	---	---	---

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. This test to be performed at the conclusion of the holding period.
3. With a metal plate shorting the line the position of the voltage standing wave minimum shall be determined. The metal plate shall be replaced by the tube and the position of the voltage standing wave minimum of the flat portion of the pulse shall be measured.
4. The ignitor current shall not be adjusted during life test. Life-test end points shall be measured using a fixed voltage and resistor.
5. After this test, the following requirement shall be met: Recovery time (Method 4471).
6. This revision supersedes the supersession data contained in Specification Sheet MIL-E-1/1709, dated 11 November 1971.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 80
DSA - ES

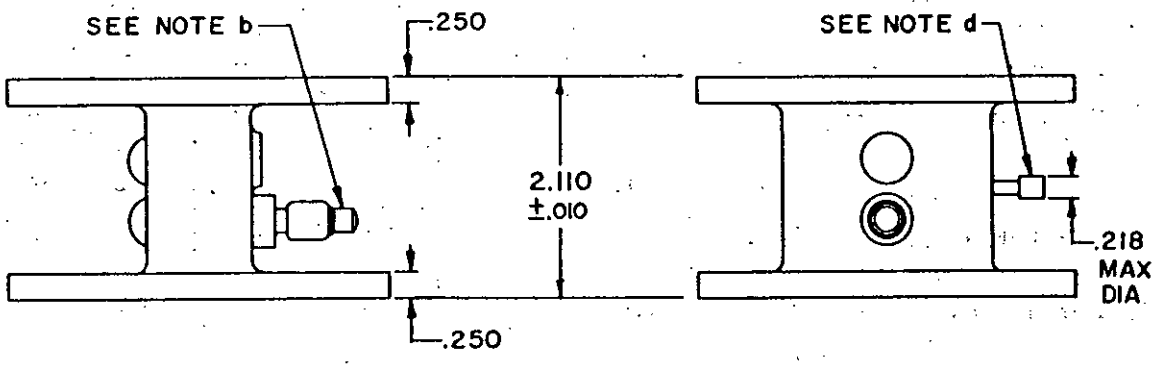
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

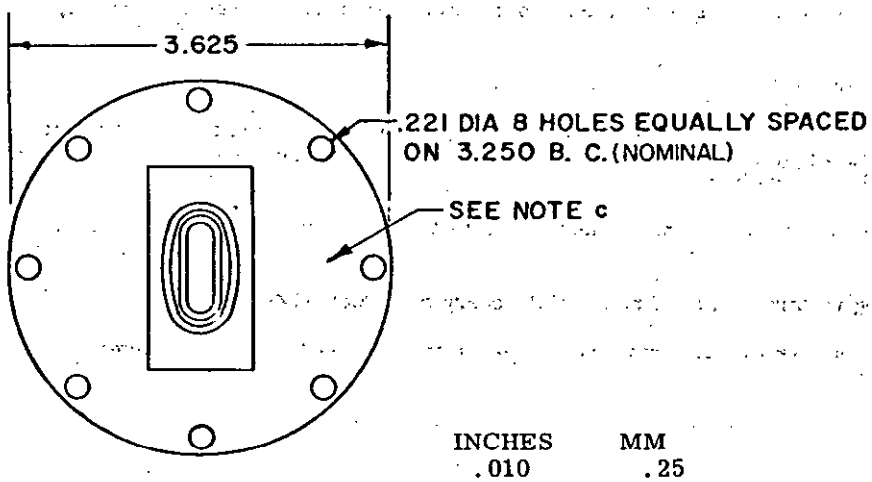
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2823)

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INCHES	MM
.010	.25
.218	5.54
.221	5.61
.250	6.35
2.110	53.59
3.250	82.55
3.625	92.08

NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. Skirted miniature cap C1-3. Cap not to extend beyond edge of flange.
c. Rhodium flash over silver plating optional.
d. Exhaust tube not to extend beyond edge of flange.
(E) e. Unless otherwise specified, tolerance is ±.016 (.41 mm).

FIGURE 1. Outline drawing of electron tube type 5865.

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